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COMMON QUARTERLY EXAMINATION - 2025

Standard XII

Reg.No.

CHEMISTRY

Time : 3.00 hrs

Marks : 70

Part - I

15 x 1 = 15

I. Answer all the questions.

1. In the electrolytic refining of copper, which one of the following is used as anode?
   a) pure copper b) impure copper c) carbon rod d) platinum electrode
2. Which of the following is not sp^2 hybridised?
   a) graphite b) graphene c) fullarene d) dry ice
3. The stability of +1 oxidation state increases in the sequence,
   a) Al < Ga < In < Tl b) Tl < In < Ga < Al
   c) In < Tl < Ga < Al d) Ga < In < Al < Tl
4. The compound used as high energy fuel for propellant is
   a) BF_3 b) B_2H_6 c) $AlCl_3$ d) H_3BO_3
5. H_2SO_5 is called as
   a) sulphurous acid b) dithionus acid c) caro's acid d) Marshall's acid
6. The crystal has metal excess defect,
   a) FeO b) AgCl c) NaCl d) $CdCl_2$
7. For the reaction, $2NH_3 \rightarrow N_2 + 3H_2$, if $\frac{-d[NH_3]}{dt} = k_1[NH_3]$, $\frac{d[N_2]}{dt} = k_2[NH_3]$, $\frac{d[H_2]}{dt} = k_3[NH_3]$, then the relation between k_1, k_2 and k_3 is
   a) $k_1 = k_2 = k_3$ b) $k_1 = 3k_2 = 2k_3$ c) $1.5k_1 = 3k_2 = k_3$ d) $2k_1 = k_2 = 3k_3$
8. What is the pH of 0.1 M CH_3COOH solution? ($Ka = 1.8 \times 10^{-5}$)
   a) 2.87 b) 3 c) 1.8 d) 0.1
9. Most easily liquifiable gas is
   a) Ar b) Ne c) He d) Kr
10. Which of the following oxidation state is most common among the lanthanoids?
    a) +4 b) +2 c) +5 d) +3
11. Ethanol $\xrightarrow{PCC}$ X. X is
    a) acetaldehyde b) propanal c) 1-butanol d) 2-butanol
12. Which of the following compounds on reaction with methyl magnesium bromide will give tertiary alcohol,
    a) benzaldehyde b) propanoic acid c) methyl propanoate d) acetaldehyde
13. Which one of the following reduces Tollens reagent?
    a) formic acid b) acetic acid c) benzophenone d) none of these
14. In which of the following reaction new carbon-carbon bond is not formed?
    a) Aldol condensation b) Friedel-Crafts reaction
    c) Kolbe's reaction d) Wolf Kishner reaction
15. On hydrolysis PCl_3 gives
    a) H_3PO_3 b) PH_3 c) H_3PO_4 d) $POCl_3$

Part - II

II. Answer any 6 questions. (Q.No.24 is compulsory) 6 x 2 = 12

16. Give the limitations of Ellingham diagram.

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17. Give one example for each :
    a) Chalcogen b) Icosagen c) Pnictogen d) Tetragen
18. What are inter-halogen compounds? Give examples.
19. Which is more stable : Fe^{2+} or Fe^{3+}. Why?
20. Define solubility product.
21. Convert : Glycol into Acetaldehyde
22. Explain Kolbe's reaction.
23. What is Williamson synthesis?
24. Find the order of the following reactions :
    a) Radioactive disintegration of $_{92}U^{238}$
    b) $2A + 3B \rightarrow$ Products. rate = $k[A]^{1/2}[B]^2$

Part - III

III. Answer any 6 questions. (Q.No.33 is compulsory) **6 x 3 = 18**

25. Write a short note on anamolous properties of the first element of p-block.
26. Write the uses of Xenon.
27. What is Ziegler-Natta catalyst? Give its uses.
28. Give the differences between order and molecularity.
29. What is common ion effect? Explain with example.
30. What is Glycerose? How is it formed?
31. Explain the coupling reaction of phenol.
32. What is Perkin's reaction?
33. Barium has body centered unit cell with a length of 508 pm along an edge. What is the density of the Barium in $g\,cm^{-3}$?

Part - IV

IV. Answer all the questions. **5 x 5 = 25**

34. a) i) Distinguish between mineral and ore? (3)
        ii) Give any two uses of zinc. (2) ***(OR)***
    b) i) What is ethyl borate test? (3)
        ii) Give two uses of potash alum. (2)
35. a) i) Explain the reaction of dehydrating nature of sulphuric acid. (3)
        ii) State Hume-Rothery rule. (2) ***(OR)***
    b) Distinguish between crystalline solids & amorphous solids. (5)
36. a) Derive integrated first order equation. (5) ***(OR)***
    b) i) Define pH. (2)
        ii) What is ionic product of water? Give its value at room temperature. (3)
37. a) i) Convert : Glycerol into Acrolein. (2)
        ii) What is the action of con.H_2SO_4 on Glycol. (3) ***(OR)***
    b) Write any three electrophilic substitution reaction of phenol. (5)
38. a) Explain the mechanism of aldol condensation. (5) ***(OR)***
    b) An organic compound A with molecular formula C_7H_6O reduces Tollens reagent, not reduces Fehlings solution. A reacts with 50% NaOH to give B and C. C on treatment with sodalime to give D. Identify A,B,C & D. Write the equations. (5)

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